INCH-POUND

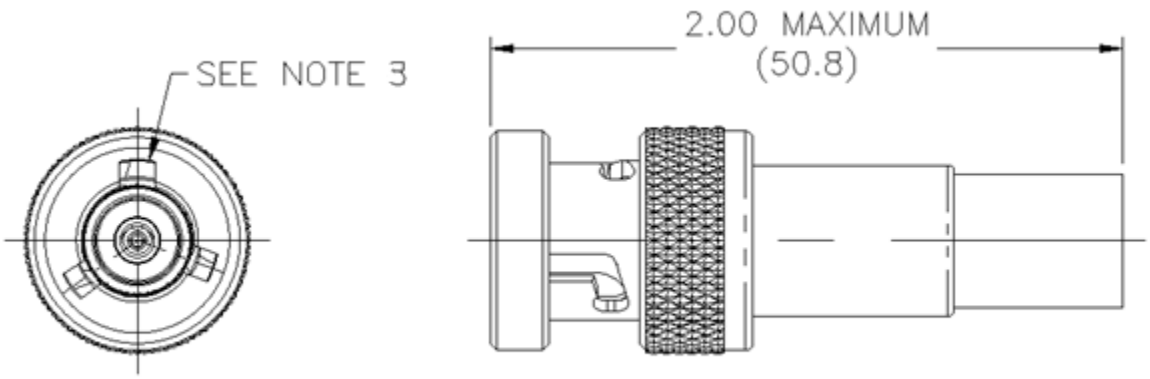
MIL-PRF-49142/14A
23 August 2023
SUPERSEDING
MIL-PRF-49142/14
4 December 2019

PERFORMANCE SPECIFICATION SHEET

CONNECTOR, PLUG, ELECTRICAL, TRIAXIAL, MIL-STD-1553 DATABUS,
(SERIES TRK, GENDER CHANGEABLE CONTACT , CLASS 2)

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of
this specification sheet and MIL-PRF-49142.

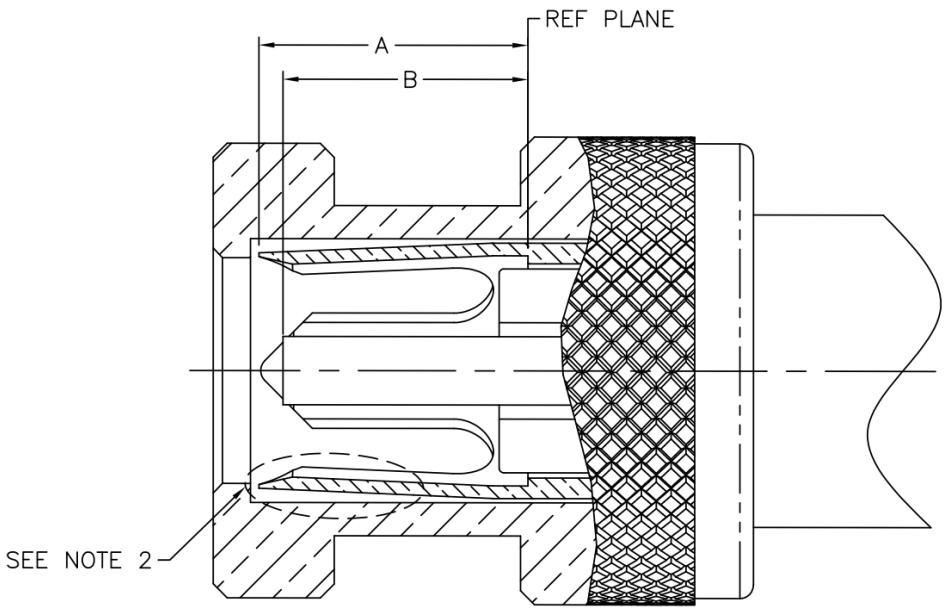



NOTES:
1. Dimensions are in inches.
2. Undimensioned pictorial representations for reference only.
3. Multiple keyring configurations.

FIGURE 1. General configuration.


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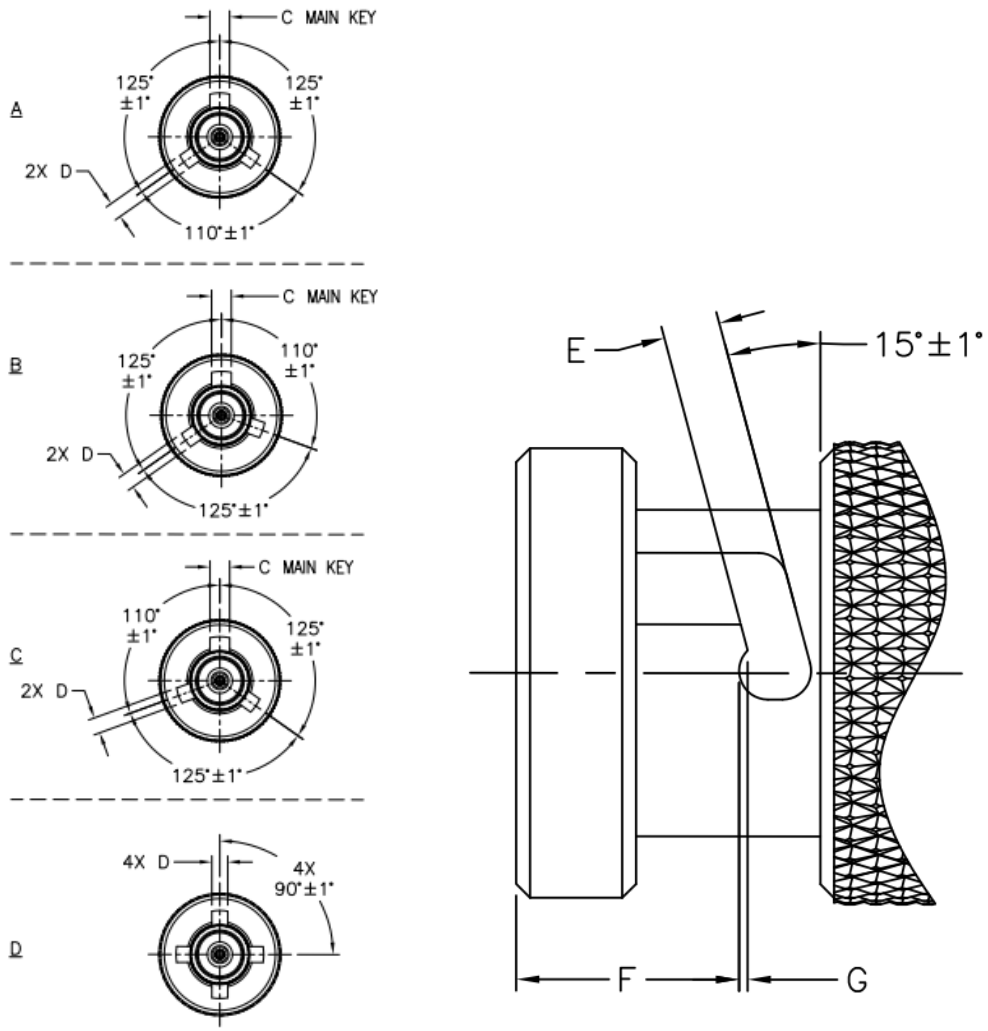



| LTR | Inch es | |
|---|---|---|
| | Min | Max |
| A | .281 | .303 |
| B | .248 | .319 |

NOTES:
1. Dimensions are in inches.
2. Slot design of alignment fingers optional.
3. See Figure 3 for keying configurations.
4. See figures 4 and 5 for signal contact polarity options

FIGURE 2. TRK plug interface dimensions.


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| LTR | Inches | |
|---|---|---|
| | Min | Max |
| C | .077 | .083 |
| D | .056 | .068 |
| E | See keying detail | |
| F | .211 | -- |
| G | .015 | -- |

NOTES:
1. Dimensions are in inches.

FIGURE 3. Keyring configurations.



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ENGINEERING DATA:

Nominal impedance: Non-constant.

Signal range: Databus - MIL-STD-1553 databus frequency range.
RF Signaling - as required by the applicable system requirements.

Voltage rating:

At sea level: 400 V rms maximum working voltage.

At 70,000 feet (4.437 kPa): 100 V rms maximum working voltage.

Operating temperature -65°C to +125°C.

Storage temperature: -65°C to +125°C.

REQUIREMENTS:

Dimensions and configuration: See Figure 1 and Figure 3.

Mating connector in accordance with MIL-PRF-49142/13.

Materials:

Insulators: PTFE fluorocarbon in accordance with ASTM D4894 or D4895, or POM in accordance with ASTM D6778.

Alternate insulators: ETFE per ASTM D3159 or PES that meet the requirements and pass the tests in Table I.

Key material properties of Polyethersulfone (PES):

Table I.

| Characteristics | Typical Value | ASTM Test Method |
|---|---|---|
| Specific Gravity | 1.37 | D792 |
| Water Absorption, 24hrs @ 23 °C | 0.54 % | D570 |
| Tensile Strength | 12000 psi (83 MPa) | D638 |
| Tensile Elongation | 30 % | D638 |
| Tensile Modulus | 0.39 x 106 psi (2689 MPa) | D638 |
| Dielectric Strength | 380 V/mil (15.0 KV/mm) | D149 |
| Deflection Temperature @ 264 psi (1820 KPa) | 398 °F (203 °C) | D648 |

Connector bodies: Brass in accordance with ASTM B16/B16M, ASTM B36/B36M or ASTM B121/B121M. Or copper beryllium in accordance with ASTM B194, ASTM B196/B196M or ASTM B197/B197M

All other components: In accordance with MIL-PRF-49142.



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Force to engage and disengage:

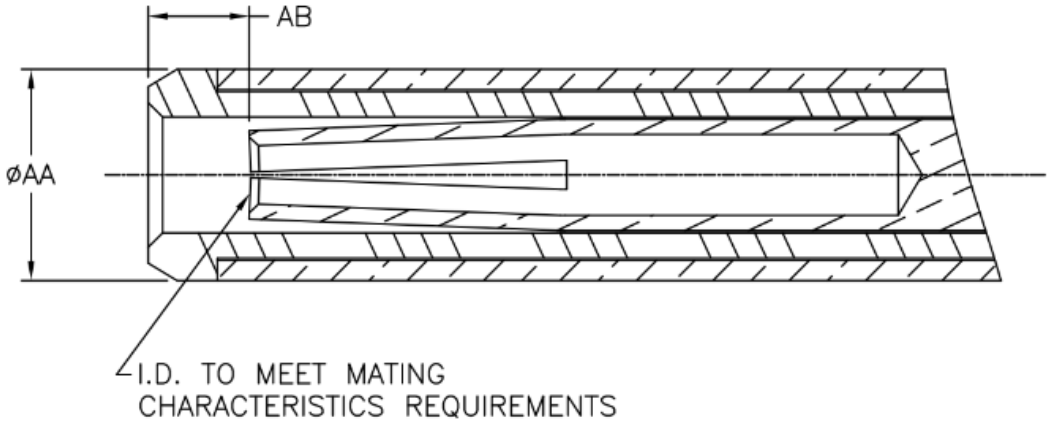
Longitudinal force: 3 pounds (13.34 N) maximum.

Torque: 2.5-inch pounds (.28 Nm) maximum.

Coupling proof torque: Not applicable.

Mating characteristics: See Figure 2, Figure 4 and Figure 5 for dimensions and force testing.

Signal Contact, Pin:




| LTR | Inches | |
|---|---|---|
| | Min | Max |
| AA | .072 | .074 |
| AB | .040 | .061 |

FIGURE 4. Pin Interface dimensions.

Center contact (socket):

Steel test pin finish: 16 micorinches (.406 µm).
Insertion depth: .050 inches (3.17mm) minimum.
Oversize test pin: .025 inches (.635 mm) diameter minimum.
Number of insertions: 1.

Insertion force test:

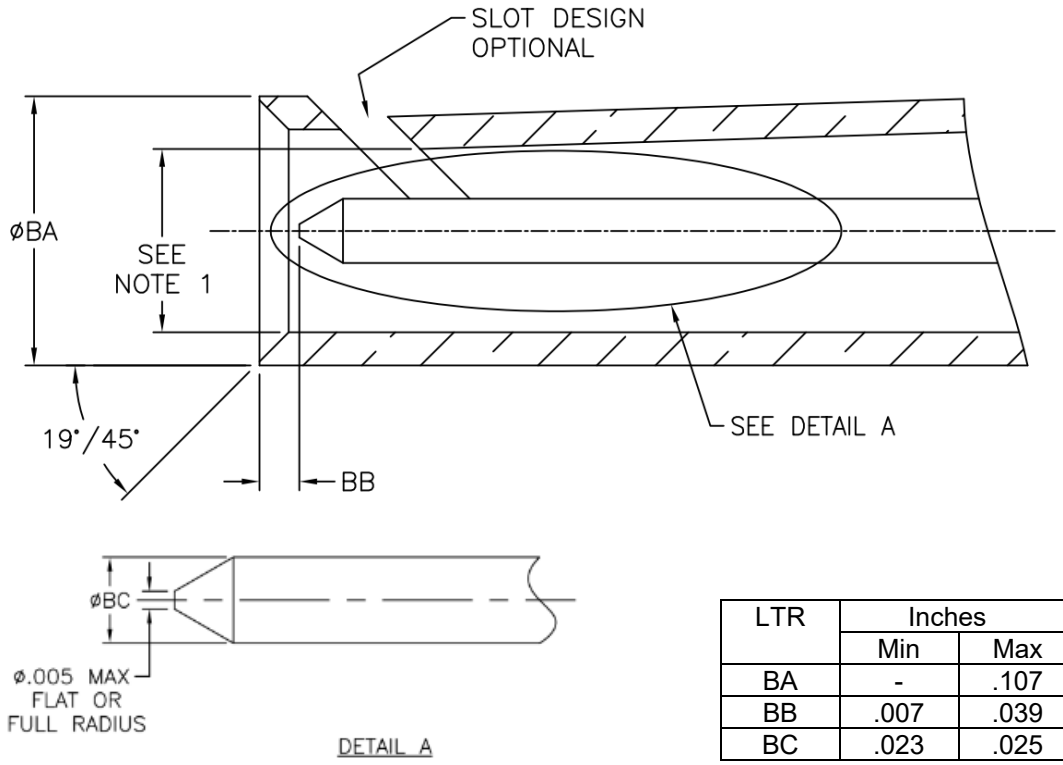
Steel test pin diameter: .0245 inches (.622 mm) minimum.
Insertion force: 1 pounds (4.45 N) maximum.

Withdraw force test:

Steel test pin diameter: .0235 inches (.597 mm) maximum.
Withdrawal force: 0.5 ounces (.14 N) minimum.


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Signal Contact, Socket:




| LTR | Inches | |
|---|---|---|
| | Min | Max |
| BA | - | .107 |
| BB | .007 | .039 |
| BC | .023 | .025 |

Note 1: Set to meet the engagement and separation force.

FIGURE 5. Socket Interface Dimensions.

Intermediate contact (socket):

Steel test pin finish: 16 microinches (.406 µm).
Insertion depth: .125 inches (3.17mm) minimum.
Oversize test pin: .075 inches (1.91 mm) diameter minimum.

Number of insertions: 1.

Insertion force test:

Steel test pin diameter: .074 inches (1.88 mm) minimum.
Insertion force: 1.25 pounds (5.56 N) maximum.

Withdraw force test:

Steel test pin diameter: .072 inches (1.83 mm) maximum.
Withdrawal force: 2.0 ounces (.56 N) minimum.



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Permeability: Applicable.

Hermetic seal: Not applicable.

Leakage: Not applicable.

Insulation resistance:

Center conductor to intermediate conductor: 5,000 megohms minimum.

Intermediate conductor to outer conductor: 5,000 megohms minimum.

Conductor retention:

Center conductor retention: 2 pounds (8.90 N) minimum axial force.

Intermediate conductor retention: 6 pounds (26.69 N) minimum axial force.

Dielectric withstanding voltage at sea level:

Center conductor to intermediate conductor: 900 V rms minimum.

Intermediate conductor to outer conductor: 900 V rms minimum.

Salt spray (corrosion): Applicable.

Vibration: Applicable.

Shock: Applicable.

Thermal shock: Applicable.

Moisture resistance: Applicable.

Conductor resistance: In milliohms, maximum.

| | Initial | After environment |
|---|---|---|
| Center conductor: | 6.0 | 8.0 |
| Intermediate conductor | 3.0 | 5.0 |
| Outer conductor | 5.0 | 8.0 |

Corona level: Not applicable.

RF high potential withstanding voltage: Not applicable.



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Cable retention force (for cable outside diameter):

.154 inch (3.93 mm) and smaller: Not applicable

.155 - .189 inch (3.94 – 4.82 mm): 10 pounds (44.48 N) minimum.

.190 - .229 inch (4.83 – 5.83 mm): 20 pounds (88.96 N) minimum.

.230 - .249 inch (5.84 – 6.34 mm): 30 pounds (133.45 N) minimum.

.250 inch (6.35 mm) and larger: 40 pounds (177.93 N) minimum.

Coupling mechanism retention force: 100 pounds (444.82 N) minimum.

Rise time degradation:

Twin conductor cable: 100ns when tested in accordance with MIL-STD-1553 paragraph 4.5.2.1.1.2.

Triaxial cable: Applicable.

NOTE: Triaxial cable approved for use in RF applications.

Connector durability: 500 cycles minimum at 12 cycles per minute maximum.

RF leakage: Not applicable.



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Part identifying number (PIN): M49142/14- X X X X - XX - X

Body material finish from Table 2----------
Keying from Figure 3 ----------
Configuration from Table 3 ----------
Gender from Figure 4 or 5 (P=Pin, S=Socket) ----------
Dash number from Table 4 ----------
Termination Type from Table 5 ----------

Table 2. Body material finish and finish.

| Part identifying number (PIN): M49142/14- AXXX-XX-X | |
|---|---|
| 0 | silver plated brass |
| 3 | Not applicable |
| 4 | tin plated brass |
| 5 | tin/nickel plated beryllium copper |
| 6 | tin plated beryllium copper |
| 7 | nickel plated brass |
| | |

Table 3. Configuration.

| Part identifying number (PIN): M49142/14- XXAX-XX-X | |
|---|---|
| 1 | Cable |
| 2 | Not applicable |
| 3 | Not applicable |
| 4 | Not applicable |
| | |

Table 4. Dash Number.

| Part identifying number (PIN): M49142/14- XXXX-AA-X | |
|---|---|
| 00 | Not applicable |
| 01 | M17/176-00001 |
| 02 | 10602 or 10612 Raychem |
| 03 | TWC-78 Trompeter |
| 04 | 10613 Raychem |
| 05 | 10614 Raychem |
| 99 | Per contract |
| | |

Table 5. Termination Type.

| Part identifying number (PIN): M49142/14- XXXX-XX-A | |
|---|---|
| C | Crimping |
| S | Pre-installed Soldering |
| | |

NOTE: Components for Type C and Type S are intermateable but not interchangeable.



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Referenced documents. In addition to MIL-PRF-49142, this document references the following:

| | | |
|---|---|---|
| MIL-PRF-49142/13 | ASTM B121/B121M | ASTM D4894 |
| MIL-STD-1553 | ASTM B194 | ASTM D4895 |
| M17/176 | ASTM B196/B196M | ASTM D6778 |
| ASTM B16/B16M | ASTM B197/B197M | ASTM D3159 |
| ASTM B36/B36M | | |

## CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2023-047)

Review activities:
Army - AR, AT
Navy - AS, MC, OS
Air Force – 19

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